Image Sensing

Image-Based Sensing of Leukonychia for Early Diagnosis of Anemia Using a Smartphone Application

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Abstract—Mobile health has gained significant attention due to its low cost, portability, and ease of access. Given that more than 27% of the population suffers from iron deficiency anemia, an ample amount of research is being done to find point-of-care ways to diagnose this disease. The traditional methods are invasive and time-consuming. In this letter, we demonstrate a smartphone application that employs image processing techniques like contour and blob detection, morphological opening and closing, and pixel analysis to detect anemia. An algorithm for detecting leukonychia, which is the appearance of white spots or streaks, is developed; leukonychia is an indicator of iron deficiency anemia, and this is detected using an image analysis technique. We can detect anemia with an accuracy of 89% just by analyzing the photograph of nails from an individual; one can diagnose the early onset of anemia. This method demonstrates a low margin of error and high sensitivity of 96% compared to prior arts indicating its usability as an onsite detection technique. The method is employed in available datasets illustrating 89% accuracy overall. Furthermore, this technique is available in the form of an application and hence is very affordable and large-scale deployable in rural set-ups as well.

Index Terms—Electromagnetic wave sensors, anemia, blob detection, leukonychia, smartphone.

I. INTRODUCTION

Affecting more than two billion people worldwide, anemia is a very commonly occurring disease, and work is being done in many fields to diagnose it at its early stages. While smartphone-based telehealth has the potential to change the way in which diagnosis is delivered, being noninvasive in nature, they are yet to reach the level of accuracy of blood-based testing before it can replace them. The work done in this letter is along the same lines. It does not aim to replace medical diagnostics but rather to serve as a sort of preliminary test to make individuals aware of their health status. The burden of blood testing coupled with the additional cost of equipment makes telehealth an appealing alternative compared to its competitors [1], [2], [3].

In this letter, we detect the possibility of a patient having anemia through the phenomenon of leukonychia, which is the appearance of white spots or streaks in such patients [1], [4]. The detection of this disease does not employ color detection like most image processing algorithms where blood hemoglobin levels are tested for [5]. Instead, detection is done by an edge-object detection process where all the white spots in the nail are isolated (considered to be blobs) and counted. Image processing techniques like contour and blob detection, morphological opening and closing, and pixel analysis have been employed to detect iron deficiency anemia [6]. Based on the nature of its presence, a prediction is made on whether the user has anemia. The output image is analyzed, and a histogram is prepared. A threshold has been set after a thorough investigation and literature review [5], [7]. If the sum of histogram bins crosses the threshold, the user may have anemia. Prediction is made with an accuracy of 89% and sensitivity of 96%, which is high compared to other methods and work done in this area.

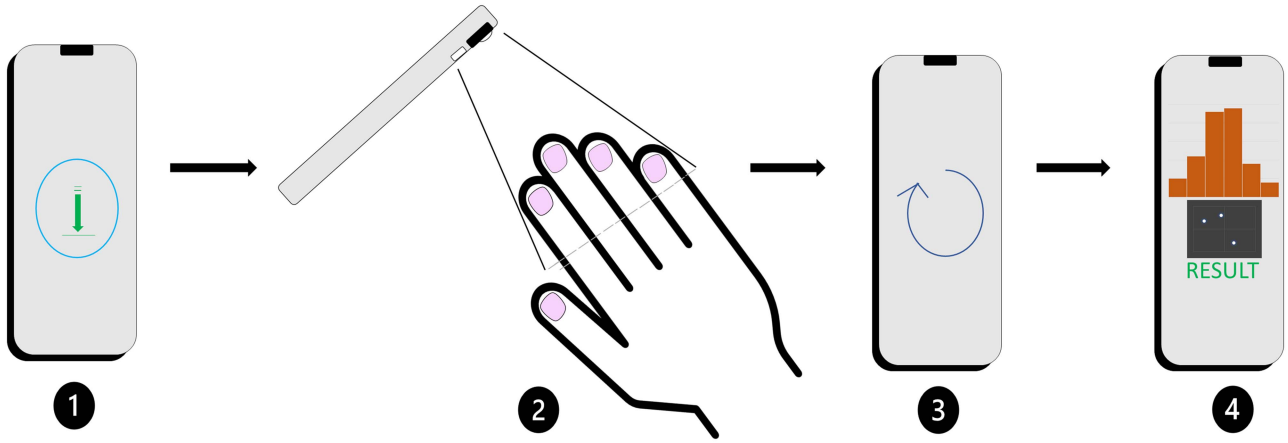
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Estimating hemoglobin levels by analyzing the color and metadata of fingernail beds is one method that has been attempted to detect anemia. This is one of the intuitive approaches to the problem; however, this method relies on the visual intensity of red visible in nail beds at any given time, which is a very dynamic factor [8], [9]. While being more accurate than applications like HemaApp, which performs the same test but through the finger and is susceptible to skin tone, it still suffers from the issue of variability of blood flow. This method is suitable for determining parameters like the instantaneous heart rate of an individual or oxygen levels, which has been explored by the authors of this letter in a separate work. Another parameter considered was the nature of respiration in individuals through sound processing algorithms; however, sporadic breathing could be an indicator of many diseases and is not exclusive to anemia [10]. Even skin color using OpenCV to detect paleness or yellowish skin has been considered, but the accuracy was very low [11]. Blood samples are often analyzed by image processing techniques, and those have been translated into the application developed. Combined with the agnostic nature of fingernail beds to interference and leukonychia being a well-known diagnosis parameter [12], the authors of this letter have developed an algorithm, implemented in MATLAB mobile, for early diagnosis of anemia (see Fig. 1). The novelty introduced in the project is that by utilizing all available features and sensors [13] in a smartphone, we can measure the degree of leukonychia [14], [15] for iron deficiency anemia. The ease of use, which involves only clicking an image of the nails, means that anyone can take this test at any time. The processing involved is moderate enough that any smartphone can run it without taxing the system.

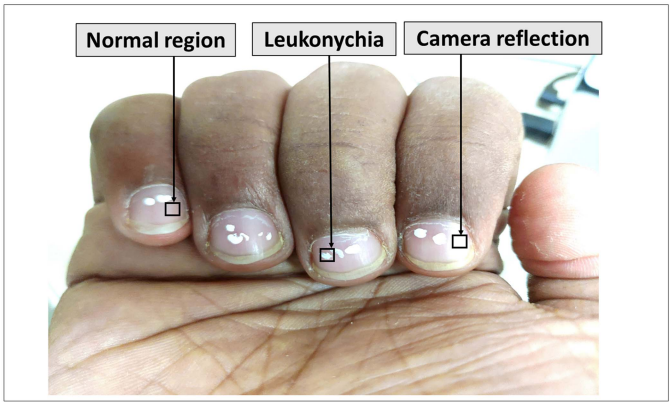
II. METHODOLOGY

This technique has the potential to be used by the general public or at-risk populations as a noninvasive anemia self-screening tool. Smartphone-based anemia testing is done with a single photograph and

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(a)



(b)

Fig. 1. (a) Implementation of application for detection of anemia. ① Installation of the application. ② Open the application and take a photograph of fingernail beds. ③ Processing the image. ④ Results are displayed. (b) Different features in an image under consideration.

no customized calibration procedure. It is well known that fingernail beds have very little melanin in comparison to other regions of the skin, making this procedure indifferent to skin tone. The major component of the algorithm [16], [17] is blob detection to choose regions of interest from each finger. Each person's color data is collected from each location and averaged across all their fingers. Due to the limited color diversity between different fingers, this method was found to be acceptable. Using MATLAB mobile, picture metadata, fingernail data, and skin color data were retrieved from smartphone photographs of the fingernail bed. To guarantee that fingernail abnormality was not included in the analysis, regions of interest from which fingernail and skin color data were taken were carefully chosen. These zones of interest, which were 500×500 pixels^2 in size and corresponded to around 10 mm^2 of the fingernail, were chosen from each finger, omitting the thumb.

The image is first converted to grayscale to differentiate the white spots from the rest of the image. A histogram is plotted for the entire dataset, and a threshold gray level is observed ($>$240 on a scale from 0 to 250 in this case) in all the pictures. This value is used to make a mask around all points that exceed the threshold. We set a minimum blob size of 50 pixels, and the resulting image is then processed by edge detection and morphological operations—masking and opening to zero on the white spots and streaks.

Finally, we apply blob detection to identify the number of white spots visible (see Fig. 2). Five is taken as the threshold blob count, which is obtained by testing the algorithm with values from 1 to 20, plotting the receiver-operating-characteristic (ROC) curve, and identifying the threshold that corresponds to the optimal cut point (highest sensitivity and specificity). Any image with a blob count greater than five can be considered a possible case of anemia. This threshold ensures that many false positives are not obtained, and a significant number of white spots are present to indicate leukonychia. The photographs frequently have overlapping or extremely close-together regions. Due to their unique shapes, the issue of evaluating overlapped spots becomes complicated. We perform precise diagnosis via watershed segmentation.

Using the procedure above, the algorithm correctly detected iron deficiency anemia in the 286 images provided from various sources such as Medical News Today, NCBI website, datasets from Kaggle, and research papers that include different ethnicities, genders, and age groups, with an accuracy of 89%. A caveat in the process is that the image should have a certain level of brightness to successfully identify the white spots, which is ensured by using the camera flash. A summary of the algorithm is given in Fig. 3.

III. RESULTS

Iron deficiency is one of the most typical causes of anemia, whereas other important vitamins can also contribute to anemia [18]. The likelihood that anemia will be discovered is decreased because the symptoms might not be evident unless the anemia is severe. The

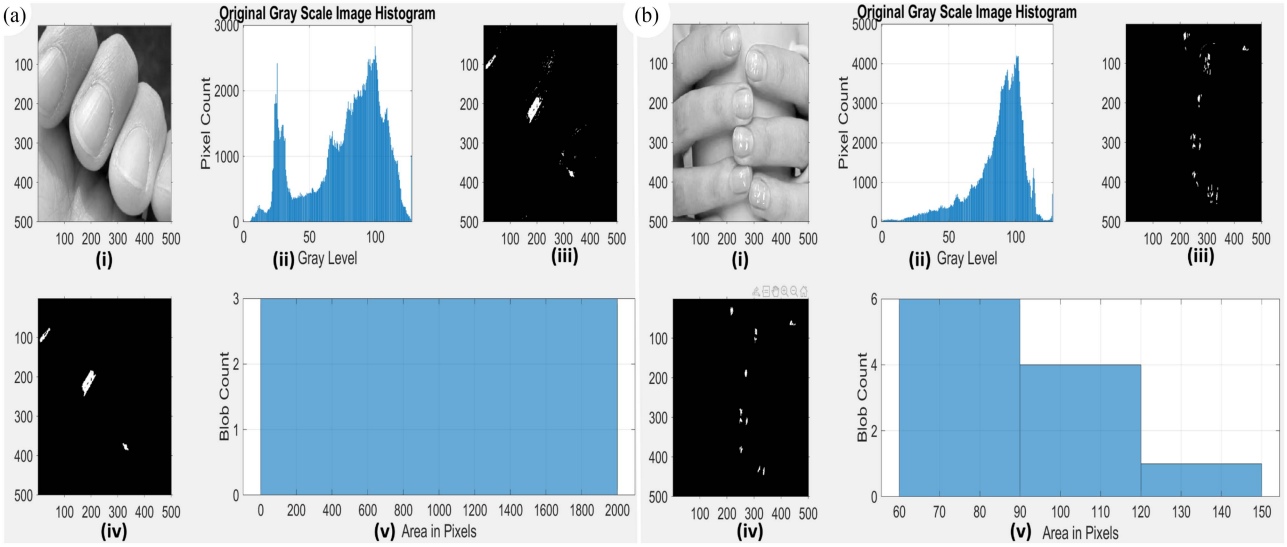


Fig. 2. Output of algorithm. (a) Nails of a normal person and output with blob count <5. (b) Image of an anemic patient with blob count >5. In both cases, (i) import image, (ii) perform gray-level thresholding, (iii) masking and morphological operations, (iv) watershed segmentation, and (v) blob count.

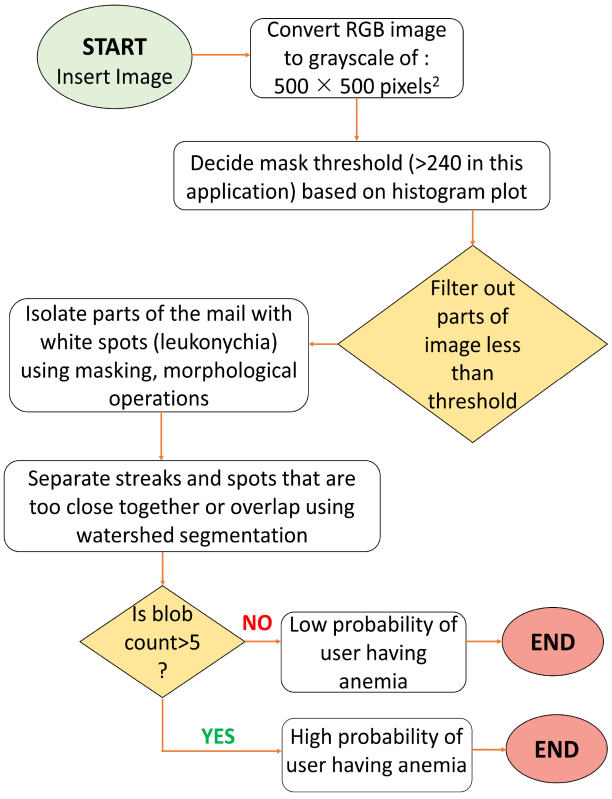


Fig. 3. Algorithm implemented in the anemia detection application. All the processes and decisions taken have been shown.

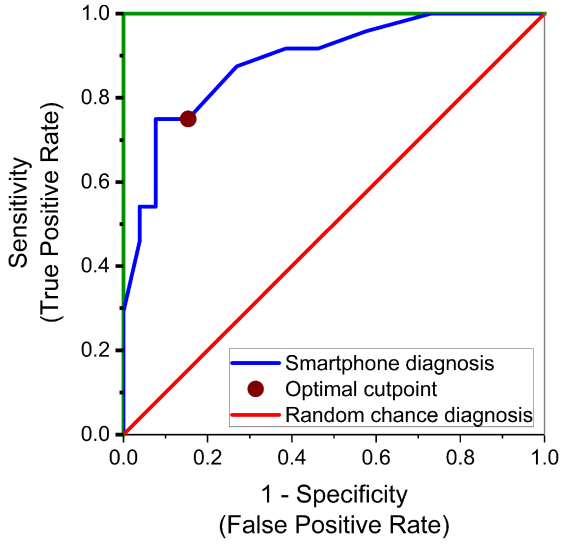


Fig. 4. ROC analysis.

two-tailed Student's t-test was used to evaluate statistical significance ($p < 0.05$) under the assumption of unequal variance in R. The efficient-score approach is used to calculate the sensitivity and specificity, giving a 96% confidence interval.

We were able to predict the occurrence of iron deficiency anemia with an accuracy of 89%, as depicted in Fig. 4 by the ROC curve. The algorithm employed was successful in detecting almost all the white spots and streaks visible while only failing on occasions where there was overlap between adjacent spots or very close ones, that too quite rarely. The watershed segmentation employed for separation could be substituted with an alternate function or a custom-made one, but in our case, it provides satisfactory results. The findings during the work demonstrated that using flash during photography was not necessary (but is recommended) to produce photographs suitable for detecting anemia. Additionally, RAW photographs offer information straight from the camera sensor [19], [20] without standard processing or reduction seen in popular formats like JPEG [2]. Inconsistent quality throughout the image and nonuniform lighting resulted in a few false positive or false negative outcomes. However, it is observed that the flashlight in the camera provides uniform illuminance across the image and significantly improves the performance (~40% increase) by avoiding any issues due to environmental lighting. Errors due to background texture have been removed by implementing the global thresholding function. A high sensitivity means that fewer false negative results occur, which reduces the number of disease cases that are overlooked.

Table 1. Comparison of different methods invoked in anemia detection.

S. no.	Method	A (%)	S (%)	Ref.
1.	**Blob-based detection**	**89**	**96**	**This work**
2.	Hemoglobin detection (nails)	88	92	[1]
3.	Hemoglobin detection (finger)	76	87	[1]
4.	Blood sample	95	97	[6], [15]
5.	Ocular Images	71	89	[2], [21]
6.	Breathing pattern	51	9	[7]
7.	Paleness of skin	71	84	[3], [4], [9]

A = Accuracy, S = Sensitivity.

It does not directly indicate that a method is more reliable, as there is also an increased chance of false positives. A combined analysis of accuracy and sensitivity suggests the reliability of a technique. Table 1 shows a comparison of all existing detection methods. Among those in which different parameters are tested, it is the best, being only beaten by a blood test.

Advancements in image quality and algorithmic corrections can improve anemia prediction. To make a more precise and reliable prognosis, additional factors (such as hemoglobin, pallor, and irregular breathing) can be added to a single reading. The data with which we can check the anemia detection feature is currently limited, and the algorithm developed has been optimized for images of a single hand. However, two-hand imaging can also be used, but a bigger set of data having two-hand images are needed to validate. Extensive real-life testing should also be done to overcome inaccuracies [22]. Fig. 4 graphically depicts the algorithm's diagnostic performance in comparison to a perfect diagnostic (green lines) and a random chance diagnosis (red line), with an area under the curve of 1 for the former and 0.5 for the latter. The area under the curve of 0.89 is observed in the case of this noninvasive smartphone app. This suggests that this algorithm can perform valid diagnostics.

IV. CONCLUSION

An algorithm implemented in MATLAB mobile is developed for the detection of iron deficiency anemia using the analysis of smartphone images. Given its effectiveness and the widespread occurrence of anemia in the world, this noninvasive technology—which only requires images taken with smartphones—has important implications as a widely available screening tool for both at-risk populations and the general public. The image sensing algorithm demonstrates 89% accuracy in detection with the blob analysis technique. This method also shows 96% sensitivity alongside a low margin of error. This method shows significant improvement in accuracy and sensitivity over its counterparts. A notable enhancement over current point-of-care anemia screening is the capacity to detect anemia with a high degree of sensitivity at a low cost, fully noninvasively, and without the use of any extra smartphone attachments or calibration tools. The current point-of-care anemia screening systems' external equipment requirements are a significant barrier to use. Since sensitivity may accurately identify a large majority of anemia cases, it is crucial for a screening technique to be optimized. This noninvasive, affordable, patient-operated application presented in the letter enables those who are at risk of anemia to monitor their status using only the native hardware provided with their own smartphone, which is an increasingly common occurrence worldwide. This is especially important in low-resource countries, where mobile phone networks are pervasive in contrast to the relative paucity of medical infrastructure. To calculate the parameters more precisely, extra effort can be made to include elements like calibration, a larger dataset, or an AI-based learning method.

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